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Deep Unfolding Network for PAPR Reduction in Multi-Carrier OFDM Systems

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***Abstract*—In this letter, we propose a Deep Unfolding Network called PR-DUN to reduce the peak-to-average power ratio (PAPR), which is a thorny problem in Orthogonal Frequency-Division Multiplexing (OFDM) systems. The proposed multi-layer model is constructed by unrolling an iterative algorithm resulting in layers with trainable parameters, which are optimized to minimize a loss function related to the PAPR value. The deep unfolding model uses the backpropagation algorithm to transfer gradients backwards to adjust parameters. Furthermore, the proposed scheme can accommodate any transmit power constraint, and therefore can control the power increase caused by the auxiliary signal. Simulation results show that the proposed PR-DUN model achieves a larger PAPR reduction and a smaller bit error rate while being less computationally intensive than related solutions.**

***Index Terms*—Peak-to-average power ratio (PAPR), bit error rate (BER), orthogonal frequency-division multiplexing (OFDM), deep unfolding network.**

I. INTRODUCTION

In Orthogonal Frequency-Division Multiplexing (OFDM) systems, a large peak-to-average power ratio (PAPR) can cause the power amplifiers (PAs) to operate in the nonlinear regime, and therefore lead to signal distortion and data rate degradation. To suppress the PAPR, the algorithm proposed in [1] obtains a peak-cancellation signal through an iterative algorithm that relies on the Signal to Clipping noise power Ratio (SCR), which has several advantages. It reduces the PAPR without increasing signal distortions or modifying the modulation/demodulation scheme by reserving a small set of sub-carriers in the frequency domain to inject peak-cancellation signals. Further, it requires side information only if the reserved sub-carriers' positions change and it has low computational complexity as it avoids multiple Fast Fourier Transform (FFT) and Inverse FFT (IFFT) operations. However, [1] relies on a *static* threshold that does not fully exploit the power of the reserved sub-carriers to generate peak-cancellation signals, thus limiting the PAPR reduction. In addition, it does not consider the transmit power increment that results from the added auxiliary signal.

To facilitate the sub-carrier reservation, the works in [2]–[4] proposed iterative algorithms based on the least squares approximation of the clipping noise. They aim to reduce the convergence time at the expense of increasing the computational complexity. The work in [5], on the other hand, optimized the sub-carrier reservation set in the frequency domain using a genetic algorithm, and adopted the SCR algorithm to suppress peaks. However, due to the *dynamic* input signal, it is impossible for the methods in [2]–[5] to control the power. Moreover, the *static* setting of the PAPR threshold in these works limits the PAPR reduction achieved, and can even cause violations of transmit power constraints.

The works in [6]–[10] rely on neural network models, which are black-box optimization methods, to alleviate the PAPR problem. In [6], a neural network is trained to approximate the desired output signal pattern by modifying the constellation so that the PAPR is reduced. However, due to the dynamic input signal, distortion due to the modified constellation can lead to decoding errors. In [7], the encoder and decoder of an autoencoder network substitute the modulation and demodulation modules, respectively, simultaneously minimizing the PAPR and inter-symbol interference. However, when the channel conditions vary, the modulated signal suffers from an I/Q imbalance resulting in an increased bit error rate (BER) at the receiver. Similarly, to simultaneously reduce the PAPR and improve the signal reconstruction, a real-valued neural network is designed in [8]. But the I/Q imbalance is also a critical factor that challenges the robustness of this network against varying channel conditions. The works in [9], [10] generate peak cancellation signals using neural network models to exploit the advantages of the sub-carrier reservation method. The models, however, consider a static PAPR threshold while ignoring the power limitation, which limits the PAPR reduction.

We develop a low-complexity approach by *unrolling* the SCR algorithm as a multiple-layer model called the peak-reduction deep unfolding network (PR-DUN). Our proposed model adaptively generates auxiliary signals to suppress power peaks for dynamic input signals. In a nutshell, the main contributions of our work include: (i) each layer has one trainable parameter that is optimized to minimize a loss function of the peaks, so the proposed model requires very few parameters to be fine-tuned; (ii) a training algorithm is developed to train our proposed model to reduce PAPR; (iii) an algorithm for guaranteeing the power limitation is embedded into the model; and (iv) via simulation, we show that our proposed model

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achieves the largest PAPR reduction among the related works, its inference time is short, and it is robust to varying channel conditions.

II. System Model

Without loss of generality, we consider an OFDM system where a block of bits is mapped into the N-element complex-valued OFDM vector of QAM constellation points $\mathbf{X} = [X_0, X_1, \ldots, X_{N-1}]^T$, with X_k being the complex symbol located at the k^{th} sub-carrier.

$$x_n = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k e^{\frac{j2\pi nk}{N}}.$$

In the matrix form, we have the equivalent $\mathbf{x} = \mathbf{Q}\mathbf{X}$, where $\mathbf{Q}$ is the N-point IFFT matrix.

The PAPR of signal $\mathbf{x}$ is computed as the ratio of its peak power to its average power, i.e., $\text{PAPR}(\mathbf{x}) = \max_{0\leqslant n\leqslant N-1} |x_n|^2 / E\left[\|\mathbf{x}\|_2^2\right]$. To measure the PAPR reduction performance, we consider the Complementary Cumulative Distribution Function (CCDF), which denotes the probability that the PAPR exceeds a certain threshold PAPR_{th}, as follows

$$\text{CCDF}(\mathbf{x}) = \Pr\left(\text{PAPR}(\mathbf{x}) > \text{PAPR}_{\text{th}}\right).$$

The sub-carrier reservation method reserves several slots in the frequency domain for an auxiliary signal $\mathbf{c} = [c_0, c_1, \ldots, c_{N-1}]^T$ to cancel peak powers. The frequency domain of the auxiliary signal is denoted as $\mathbf{C} = [C_0, C_1, \ldots, C_{N-1}]^T$. The data and auxiliary signal are arranged in disjoint positions in the frequency domain, such that $X_k + C_k = X_k$ for $k \notin \mathcal{R}^{\text{S}}$, and $X_k + C_k = C_k$ for $k \in \mathcal{R}^{\text{S}}$. To reduce the PAPR, the auxiliary signal $\mathbf{c}$ is obtained by solving the following optimization problem

$$\min_{\mathbf{c}} \max_{0\leqslant n\leqslant N-1} |x_n + c_n|. \tag{1}$$

Since (1) is a quadratically constrained quadratic programming (QCQP) problem, the optimal solution can be obtained using a quadratic programming solver.

III. Deep Unfolding Network for PAPR Reduction

We describe the PR-DUN model, shown in Fig. 1, then show its PAPR reduction and a power control algorithm. After that, we present the training algorithm and the computational complexity analysis.

A. Network Architecture

The l-th layer of our proposed PR-DUN model[1] is given by

$$\bar{\mathbf{x}}^{(l)} = \mathbf{x} + \mathbf{c}^{(l)}, \tag{2}$$

$$n_l = \arg\max_{0\leqslant n<N} \left|\bar{\mathbf{x}}^{(l)}[n]\right|, \tag{3}$$

$$\alpha^{(l)} = f_{\text{act}}\left(w^{(l)}\sqrt{E\left[\|x\|^2\right]} + \left|\bar{\mathbf{x}}^{(l)}[n_l]\right|\right) e^{j\angle\left(\bar{\mathbf{x}}^{(l)}[n_l]\right)}, \tag{4}$$

$$\mathbf{c}^{(l+1)} = \Pi_\Omega\left(\mathbf{c}^{(l)} - \alpha^{(l)}\mathbf{p}\left[((n-n_l))_N\right]\right), \tag{5}$$

[1]The architectures of deep neural networks (DNNs) are generic and it is unclear how a relation between input and output data is established. Therefore, DNNs are usually considered as a black-box approach. In contrast, our proposed model is explicitly constructed by modeling a PAPR reduction mechanism relying on the SCR algorithm in [1].

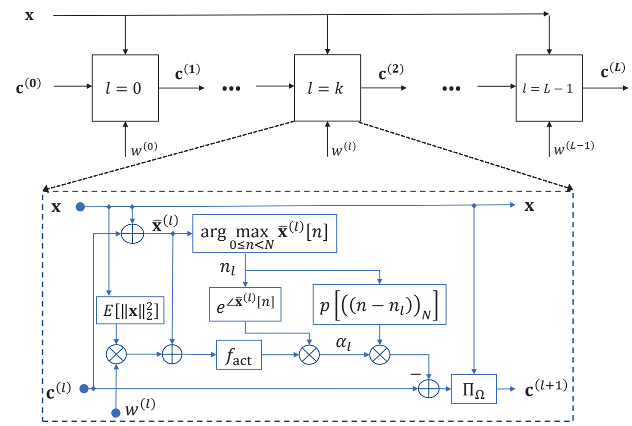

Fig. 1: Architecture of the proposed PR-DUN.

where a set of L variables $\mathbf{w} = \left\{w^{(1)}, \ldots, w^{(L)}\right\}$ is optimized using a stochastic optimization algorithm, and f_{act} is an activation function, for which we select the rectified linear unit (ReLU) that allows positive magnitude value. Moreover, $\mathbf{p} = [p_1, p_2, \ldots, p_{N-1}]^T$ presents the discrete-time domain kernel, $\mathbf{p}\left[((n-n_l))_N\right]$ is the right circular shift operator of vector $\mathbf{p}$ by n_l, and $\alpha^{(l)}$ is a scaling factor, which is updated every iteration. The kernel $\mathbf{p}$, designed such that every time the algorithm cancels a peak at position n_l of $\bar{\mathbf{x}}^{(l)}$ other positions are minimally affected by side lobes of the auxiliary signal $\mathbf{c}$, is calculated beforehand[2] [1]. The time kernel $\mathbf{p}$ is computed as $\mathbf{p} = \sqrt{N}/M\mathbf{Q}\mathbf{P}$, where the frequency domain kernel $\mathbf{P} = [P_0, P_1, \ldots, P_{N-1}]^T$. Note that $P_k = 0$ for $k \notin \mathcal{R}^{\text{S}}$, while $P_k = 1$ for $k \in \mathcal{R}^{\text{S}}$.

In (5), the operand $\Pi_\Omega(y)$ denotes the projection of y on to the feasible set Ω of signal $\mathbf{c}$. The set Ω is given as $\Omega = \left\{\mathbf{c}|E\left[\|\mathbf{x}+\mathbf{c}\|_2^2\right] \leqslant \delta^2\right\}$, where δ^2 is the power threshold. Note that the set Ω is convex, and is in fact a ball with center $\mathbf{x}$ and radius δ. Accordingly, the orthogonal projection on the set Ω is given as

$$\Pi_\Omega(\mathbf{y}) = -\mathbf{x} + \frac{\delta}{\max\left\{\sqrt{E\left[\|\mathbf{x}+\mathbf{y}\|_2^2\right]}, \delta\right\}}(\mathbf{x}+\mathbf{y}). \tag{6}$$

Our method differs from the SCR algorithm in [1] in two ways. First, (4) updates $\alpha^{(l)}$ adaptively for the l-th layer using *trainable* coefficient $w^{(l)}$ and the activation function f_{act}. Due to the auxiliary signal, the transmit power increases and can exceed the maximum allowable value. Second, to satisfy the power limitation, (5) is used to fine tune the auxiliary signal. The theoretical meaning of (6) is that there always exists a point that is the intersection between the ball Ω and a straight line crossing two points $\mathbf{x}$ and $\mathbf{y}$. We note that, when the power constraint is violated, the point $\mathbf{y}$ is outside the ball Ω. In electronic circuits, temperature fluctuation affects the PAs' operation by inducing the power gain offset, which can lead to signal clipping. To combat the varying temperature, a solution is to the estimate power gain offset first, and then reduce the

[2]FFTs and IFFTs are not required to compute the time domain kernel signal $\mathbf{p}$ every time because the reserved sub-carriers' positions in the frequency domain are fixed.

power constraint for training our model by the estimated offset. However, we are primarily presenting the proposed model in this letter, and modeling the power gain offset combined with PAPR reduction will be our next research topic.

It is also noted that the PR-DUN model is disjoint from the MAC scheduler and can be implemented in the base station's baseband software as a part of physical layer. Our proposed PR-DUN model includes one input layer, L cascaded layers, and one output layer. The signal $\mathbf{x}$ and the initial auxiliary signal $\mathbf{c}^{(0)}$ are the inputs, while the transmitted signal $\bar{\mathbf{x}}$ and modified auxiliary signal $\mathbf{c}^{(L)}$ are the outputs. The l^{th} layer is associated with variable $w^{(l)}$, which is trained to minimize a loss function described in what follows.

B. Loss Function and Batch Training

To suppress power peaks, the loss function for training the PR-DUN is the $\texttt{PRR}$ value[3], which is expressed as

$$\texttt{Loss} = \texttt{PRR}(\mathbf{x}, \mathbf{c}), \tag{7}$$

where $\texttt{PRR}(\mathbf{x}, \mathbf{c})$ is a peak reduction ratio, for signals $\mathbf{x}$ and $\mathbf{c}$, given by

$$\texttt{PRR}(\mathbf{x}, \mathbf{c}) = \frac{\max\limits_{0 \leqslant n \leqslant N-1} |x_n + c_n|^2}{E\left[\|\mathbf{x}\|_2^2\right]}.$$

The loss function for training a batch of data of size B is given by

$$\texttt{Loss}_B = \frac{1}{B} \sum_{i=0}^{B-1} \texttt{PRR}(\mathbf{x}_i, \mathbf{c}_i), \tag{8}$$

The batch training loss is used to optimize the parameter $\mathbf{w}$. Fortunately, the unfolded network allows using the backpropagation algorithm to transfer gradients backwards to optimize the parameters. Moreover, the deep unfolding network allows more flexibility in applying convex optimization.

To train our proposed model, data is generated randomly. Each data sample is a binary stream of length $(N - M)\log_2(n_{\text{QAM}})$, where n_{QAM} is the number of bits per complex symbol. Each sample passes through L layers, where each layer computes $\bar{\mathbf{x}}^{(l)}$, n_l, $\alpha^{(l)}$, $\mathbf{c}^{(l+1)}$ using (2), (3), (4), and (5), respectively. Then, $\texttt{Loss}$ is calculated for the sample based on (7). When a batch of training samples is completed, batch loss $\texttt{Loss}_B$ is aggregated using (8). The gradients are computed and back-propagated to optimize the set[4] $\mathbf{w}$. The training is repeated several times according to the number of epochs. We summarize the training algorithm in Algorithm 1.

C. Complexity Analysis

A complexity analysis of the SCR [1], SLM [11], the RNN [8], and, PRnet techniques [7] is shown in Table I. The SCR method is an iterative algorithm that generates an auxiliary signal to *soft* clip peaks. It involves several loops and a sorting algorithm, so its computational complexity is $\mathcal{O}(L\log_2 N)$[5]. The SLM algorithm's complexity, on the other hand, depends on the number of candidate phase sequences and the number of sub-carriers. In the RNN model, the number of trainable parameters depends on the hidden layer's size K, the number of the hidden layers L, the binary input size M, the output size N, and the number of parallel neural networks P. The computational complexity of the RNN model is $\mathcal{O}(PK(L + 2N))$, due to the feedfoward fully-connected architecture. The number of trainable parameters and computational complexity of the PRnet model is similar to those of the RNN model. Compared to the RNN and PRnet models, our proposed PR-DUN model requires fewer parameters to be trained. The computational complexity of the PR-DUN model is similar to that of the SCR algorithm and significantly smaller than the others.

Algorithm 1 Training Algorithm for PR-DUN

```
Set N, M, L, and δ², and select a set of reserved sub-carriers
Generate set of training samples
Initialize c and W
for each epoch do
    for each sample batch do
        for each sample x in the sample batch do
            for each layer l do
                Compute x̄^(l) by (2)
                Compute n_l by (3)
                Compute α^(l) by (4)
                Compute c^(l+1) by (5)
            end for
            Compute Loss by (7)
        end for
        Compute Loss_B by (8) and calculate gradients
        Update W via the backpropagation algorithm
    end for
end for
```

TABLE I: Complexity comparison for different techniques.

Algorithm	Complexity
SCR [1]	Computational complexity $\mathcal{O}(L\log_2 N)$
SLM [11]	Computational complexity $\mathcal{O}(MN^2)$
RNN [8]	Number of trainable parameters $P(K^2(L-2) + K + K(L-2) + 2N + 2KN + KNM)$ Computational complexity $\mathcal{O}(PK(L+2N))$
PRnet [7]	Number of trainable parameters $2(K^2(L-2) + K + K(L-2) + 2N + 2KN + KNM)$ Computational complexity $\mathcal{O}(K(L+2N))$
PR-DUN	Number of trainable parameters is only L Computational complexity $\mathcal{O}(L\log_2 N)$

IV. Simulation Results

In this section we evaluate the performance of the proposed PR-DUN model through simulations. Unless stated otherwise, we assume that the number of sub-carriers is $N = 1024$, from which $M = 32$ sub-carriers are reserved for the peak reduction purpose. Moreover, we assume 4-QAM modulation, i.e., $n_{\text{QAM}} = 2$. We generate the traing data set containing 10^4 samples from CIFAR-10 [13], created data batches with 100 samples per batch, and train the models for 10 epochs. The testing set contains 10^3 data samples. The maximum transmit

[3]Since $\texttt{PRR}(\mathbf{x}, \mathbf{c}) \geq \text{PAPR}(\mathbf{x} + \mathbf{c})$, reducing $\texttt{PRR}$ leads to a smaller PAPR.

[4]The initial values $\mathbf{w}_0$ can be computed by $\mathbf{w}_0 = (A/E[\|\mathbf{x}\|^2])^{0.5}\mathbb{1}_L$, where $\mathbb{1}_L$ is a vector of L ones, and A is a feasible PAPR threshold.

[5]$f(x) = \mathcal{O}(g(x))$ means that $\lim_{x\to\infty} |f(x)|/|g(x)| < \infty$ as in [12].

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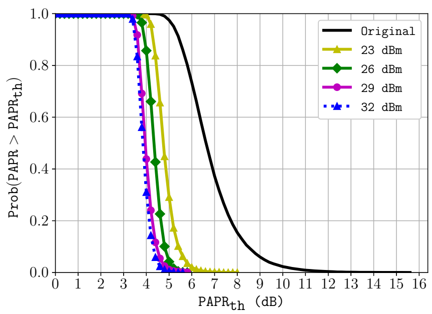

Fig. 2: PAPR reduction for different transmit power constraints δ^2.

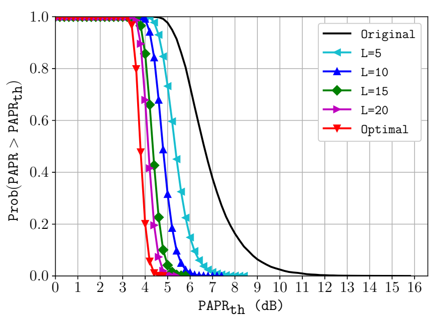

Fig. 3: PAPR reduction for different numbers of layers L.

power is set to 26dBm. The Adam algorithm is chosen to minimize the batch training loss. An experimental approach was used to determine the learning rate, which we set to 0.1 in this work. We implement the proposed algorithm using the Pytorch framework. Our experiments are conducted on an Intel(R) Xeon(R) @2.30GHz (2 CPUs), with 26GB of memory and a GPU Tesla P100-PCIE-16GB.

The PAPR performance of the PR-DUN versus the transmit power constraint δ^2 is demonstrated in Fig. 2. In this experiment, δ^2 is varied from 23dBm to 32dBm ($\delta^2 = 23, 26, 29, 32$dBm). It is shown that the PAPR reduction achieved by the proposed PR-DUN increases with the increase in δ^2. This is because more power is allocated to the auxiliary signal to cancel the peaks. However, the PAPR reduction saturates at 29dBm. This is due to the fact that the auxiliary signal itself induces secondary peaks from side lobes [1], which rise with the increase in the auxiliary signal's power.

Fig. 3 compares the PAPR reduction of the PR-DUN for different numbers of layers. We observe that increasing the number of layers in the PR-DUN results in a higher reduction in the PAPR. However, the performance saturates at a point, above which increasing the number of layers does not result in considerable performance gain. This is explained by the fact that the PAPR reduction is lower bounded by the solution of the QCQP problem, as depicted in Fig. 2.

Fig. 4 depicts the impact of the number of reserved sub-carriers on the PAPR reduction performance. It is shown that more reserved sub-carriers leads to better PAPR reduction. However, since the proposed solution is required to satisfy a transmit power constraint, the performance gain achieved by increasing the number of reserved sub-carriers inevitably

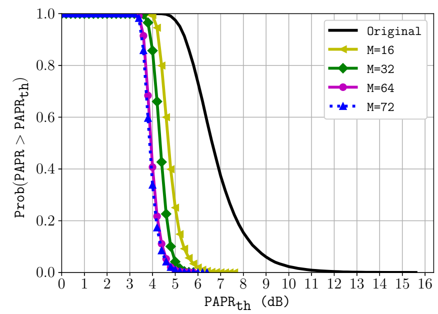

Fig. 4: PAPR reduction for different numbers of reserved sub-carriers.

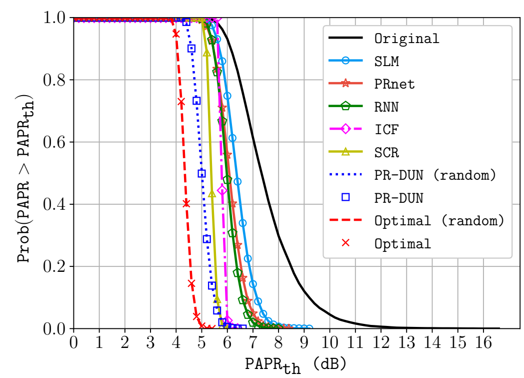

Fig. 5: PAPR reduction performance of different algorithms.

reaches a saturation point. This is because more reserved sub-carriers require more power and produce more side lobes.

In Fig. 5, we compare our 10-layer PR-DUN model to the SCR [1] and ICF [14] algorithms, which both have the PAPR target at 6dB, the SLM algorithm [11], which uses eight random phase vectors, the PR-DUN (random)[6], the RNN model [8], the PRnet model [7], and the optimal solution of the QCQP problem in (1). In Fig. 5, the number of sub-carriers is set to 2048, but the ratio between the number of sub-carriers and the number of reserved sub-carrier is fixed as $N/M = 32$. The figure shows that our proposed PR-DUN achieves the largest reduction of PAPR among the considered models and algorithms. For a 0.2 CCDF, the PR-DUN model reduces the PAPR by around 5.4dB; whereas, the SLM, PRnet, RNN, ICF, and SCR techniques reduce it by around 6.8dB, 6.2dB, and 6.1dB, 5.5dB, and 6dB respectively. Except for the ICF and SCR algorithms that satisfy their PAPR targets, compared to the SLM algorithm, the PRnet and the RNN models, our PAPR reduction is lower by at least 1.4dB. The PR-DUN outperforms the PRnet and RNN models because it induces no distortion caused by the clipping noise. The signal distortion is generated during the clipping processes of the RNN model, which hard clips the signal, and the PRnet, which adjusts the constellation. Their increased distortion limits the PAPR reduction to achieve the greater extent as their loss functions consider the BER

[6]PR-DUN (random) is our PR-DUN that allocates random sub-carrier for each data sample rather than a static one, thus providing more flexibility in resource allocation but at the cost of side information overhead [5].

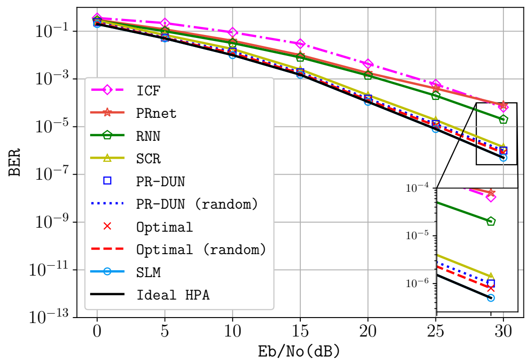

Fig. 6: BER performance of different algorithms.

reduction. Although the SLM algorithm causes no signal distortion, but its performance is limited due to the constrained number of phase vectors. In contrast, the PR-DUN model generates an auxiliary signal adaptively based on the input signal to soft clip the power peaks, which does not contribute to the signal distortion. Furthermore, the parameters of the PR-DUN model are trained to adapt to dynamic input signals to maximize its performance. Therefore, our proposed PR-DUN model achieves a better PAPR reduction than the considered related works while satisfying the power limitation.

Fig. 6 depicts the BER performance of the considered techniques under different Eb/No or signal-to-noise ratio (SNR) values. The "Ideal High Performance Amplifier (HPA)" method's data is collected by passing the original OFDM signals through a linear HPA. For the SLM method, the receiver errorlessly decodes the side information. Fig. 6 shows that when the SNR increases to 30dB, the BER of our proposed PR-DUN decreases by about 10 times lower than the ICF algorithm and the RNN and PRnet models. This is due to the signal distortion caused by the hard clipping process of the ICF algorithm and the RNN model, and the constellation adjustment of the PRnet model, make the peak-reduced signal prune to channel conditions, thus deteriorating the BER performance. In contrast, our proposed PR-DUN model uses an auxiliary signal modulated in a subset of sub-carriers to *soft* clip the power peaks, which involves neither clipping nor constellation adjustment, so that the PR-DUN model achieves a better BER performance. The PR-DUN model performs slightly better than the SCR algorithm because it achieves larger PAPR reduction than the SCR algorithm. Since the PR-DUN is trained to adaptively soft clip power peaks without consider a PAPR target, it can achieve better BER performance due to a better PAPR reduction. In addition, the PR-DUN model's BER performance is closed to that of the SLM algorithm and the Ideal HPA since it does not cause signal distortion.

Finally, we measured the PR-DUN's execution time. The average training time per epoch is approximately 15s, i.e., completing one data batch needs about 150ms. The training time includes around 24.7ms for the back propagation algorithm to update parameters and approximately 125.3ms for 100 passes of the data batch, i.e., each data sample needs roughly 1.253ms to complete. On the other hand, it takes nearly 0.934ms per sample for the PR-DUN model to execute in the inference phase, which is suitable for practical cases.

V. Conclusion

In this letter, we proposed a deep unfolding model to reduce the PAPR while satisfying the power limitation. A few trainable parameters are embedded into the proposed model to determine the clipping threshold adaptively, and the parameters are optimized over a loss function of peak values, namely the $\mathtt{PRR}$. We apply a power control algorithm to limit the power increment caused by the auxiliary signal. Simulation results show that our proposed solution achieves a larger PAPR reduction than related techniques while satisfying the power constraint and being less computationally complex. The PAPR reduction achieved by the proposed PR-DUN depends on the number of layers, number of sub-carriers, and the power limitation. Moreover, our BER performance is better than the considered neural network-based methods and iterative PAPR suppression algorithms.

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